



Correlates of specialty substance use treatment among adults with opioid use disorders



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HIGHLIGHTS

- 2488 NSDUH respondents in our sample met criteria for a past-year opioid use disorder (OUD).
- This represents 1,929,027 adults in the U.S.
- 8.3% of those with an OUD received specialty substance abuse treatment in the prior year.
- Unemployment, an arrest history, & comorbid substance use disorders were correlates of receiving treatment.

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ABSTRACT

Aims: To identify factors associated with the receipt of specialty substance use treatment among adults with opioid use disorders (OUD).

Design: Cross-sectional study based on 2010–2014 National Surveys on Drug Use and Health (NSDUH).

Setting and participants: Adults with a past-year OUD (n = 2488). The sample is representative of non-institutionalized US adults.

Measurements: Past-year OUD was determined using DSM-IV criteria. Past-year specialty substance use treatment was defined as receiving treatment for drug use at any of the following locations: rehabilitation facilities, hospitals (inpatient only), outpatient mental health centers, private doctors' offices, or methadone clinics. Multivariable logistic regression models were used to measure the independent association between potential correlates and specialty substance use treatment receipt.

Findings: Of adults with an OUD, 8.3% received past-year specialty substance use treatment. In a fully adjusted logistic regression model, the following factors were associated with increased odds of receiving specialty substance use treatment: ≥ 35 years old (adjusted Odds Ratio (aOR) = 2.55, 95% Confidence Interval (CI) = 1.04–6.26); unemployment (aOR = 1.92, 95% CI = 1.02–3.61); not in the labor force (aOR = 2.16, 95% CI = 1.15–4.06); never been married (aOR = 2.14, 95% CI = 1.04–4.39); arrested in past 12 months (aOR = 4.43, 95% CI = 2.45–7.99); opioid dependence (aOR = 3.82, 95% CI = 2.06–7.10); alcohol use disorder (aOR = 2.44, 95% CI = 1.44–4.11); and another drug use disorder (aOR = 3.22, 95% CI = 1.95–5.32). Living in a non-metropolitan county (aOR = 0.29, 95% CI = 0.12–0.68) and fair/poor health (aOR = 0.38, 95% CI = 0.17–0.86) were associated with decreased odds of receiving specialty substance use treatment.

Conclusions: These findings suggest a need for the following efforts: strategies to increase individuals' recognition of their need for OUD treatment, expansion of insurance coverage for substance use treatment, expansion of earlier intervention services, adoption of a chronic care approach to substance use treatment, and an expansion of treatment capacity for rural communities.

1. Introduction

There has been a dramatic increase in opioid use in the United

States in the last two decades. From 1999 to 2010, sales of prescription opioids in the U.S. nearly quadrupled (Centers for Disease Control and Prevention (CDC), 2011), and the prevalence of prescription opioid

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misuse and abuse increased in parallel (McHugh, Nielsen, & Weiss, 2015). These increases are thought to have contributed to a subsequent increase in heroin use as the prevalence of past-year heroin use nearly doubled between 2005 and 2012 (Substance Abuse and Mental Health Services Administration, 2013). As a result of increases in prescription opioid and heroine abuse, drug overdoses, the majority of which are opioid-related, are now the leading cause of accidental death in the U.S (Levi, Segal, & Martin, 2015).

The increase in opioid abuse in the U.S. has meant an increase in the number of individuals needing treatment (Substance Abuse and Mental Health Services Administration Center for Behavioral Health Statistics and Quality, 2015). From 2003 to 2013, the number of U.S. individuals aged 12 years or older with an opioid use disorder (OUD) increased from 1.6 million to 2.4 million (Center for Behavioral Health Statistics and Quality. Behavioral health trends in the United States, 2015). A variety of evidence-based treatments for OUD exist in the U.S., including pharmacologic treatments such as methadone, buprenorphine and naltrexone. Medication-assisted therapy (MAT), now considered to be the frontline treatment for OUD, uses pharmacologic interventions in combination with psychosocial interventions such as motivational interviewing and cognitive behavioral therapy (Schuckit, 2016). Unlike in other countries where MAT is more accessible (European Monitoring Centre for Drugs and Drug Addiction, n.d.; Merrill, 2002), U.S. federal laws and regulations limit where opioid agonist pharmacotherapies can be administered. Methadone can only be dispensed at federally certified opioid treatment programs, and buprenorphine can only be prescribed by physicians who have completed special training (Substance Abuse and Mental Health Services Administration, 2015). Despite increased demand, very few individuals in need of treatment receive MAT or other treatment. Between 2009 and 2013, only 21.5% of individuals aged 12 and older with an OUD received any treatment in the previous year (Saloner & Karthikeyan, 2015).

Although there is substantial unmet need for treatment (Becker et al., 2008; Saloner & Karthikeyan, 2015), our knowledge of factors associated with receiving treatment is limited. Barriers to accessing treatment in the U.S. include insufficient health insurance coverage, insufficient treatment capacity, and the stigma of addiction and treatment (Appel, Ellison, Jansky, & Oldak, 2004; Becker et al., 2008; Jones, Campopiano, Baldwin, & McCance-Katz, 2015). As state and federal governments in the U.S. try to address unmet treatment need for those with OUD, particularly by expanding treatment capacity, it is important to first understand who is likely to receive treatment. Doing so requires understanding the complex patient-level factors that influence treatment. Previous studies using national data have rarely focused on identifying correlates of receiving evidence-based treatments among adults with OUD. Prior studies have examined individuals with any substance use disorder (SUD) (Grella, Karno, Warda, Moore, & Niv, 2009; Harris & Edlund, 2005; Wu, Ringwalt, & Williams, 2003) or have been limited to adolescents or veterans with OUD (Finlay et al., 2016; Oliva, Harris, Trafton, & Gordon, 2012; Shiner, Leonard Westgate, Bernardy, Schnurr, & Watts, 2017; Wu, Blazer, Li, & Woody, 2011). A recent study by Wu et al. (Wu, Zhu, & Swartz, 2016) investigated correlates of receiving treatment among adults and adolescents with OUD but did not include patient-level factors, such as criminal history and employment status, that may be particularly salient for vulnerable populations.

In this study, we aimed to identify correlates of receiving substance use treatment at a specialty treatment facility among adults with OUD in the U.S. Using a large, nationally representative sample, we: 1) examined potential correlates of treatment receipt as guided by the Gelberg-Andersen behavior model of health services utilization; 2) estimated the prevalence of perceived need of treatment; and 3) evaluated reported reasons for not receiving treatment. We focused on adults since adolescents and adults with OUD are distinct populations who differ in the treatment options available to them (American Society of Addiction Medicine, n.d.) and likely face different barriers to

treatment. Unlike several previous studies (Feder et al., 2017; Saloner & Karthikeyan, 2015; Wu et al., 2016), we chose to only consider treatment received at facilities that could offer evidence-based treatments for OUD.

2. Methods

2.1. Data source

Data for this study come from individual waves of the National Survey on Drug Use and Health (NSDUH), a nationally representative, cross-sectional survey conducted annually in the continental U.S. NSDUH provides national estimates of alcohol and illicit drug use, substance use disorders, and treatment for those disorders. NSDUH samples from civilian, non-institutionalized U.S. residents aged 12 years and older. NSDUH uses multi-stage area probability sampling of dwelling units, including non-institutional group quarters such as homeless shelters, college dormitories, and group homes. Sensitive information is collected using audio computer-assisted self-interviewing (ACASI). To increase statistical power, we combined data from the five most recently available NSDUH waves (2010–2014). We did not include waves prior to 2010 to avoid introducing secular trends that might affect the results of the analysis. Weighted screening response rates ranged from 81.9% (in 2014) to 88.4% (in 2010) and weighted interview response rates ranged from 71.2% (in 2014) to 74.6% (in 2010). A more detailed description of NSDUH methodology can be found elsewhere (Center for Behavioral Health Statistics and Quality, 2014). This study was exempt from review by the University of Massachusetts Medical School Institutional Review Board because the data being used are publicly available and do not contain patient identifiable information.

2.2. Sample

This study includes adult respondents who were in need of treatment for their use of either heroin or prescription opioid pain relievers (hereafter referred to as opioids). Respondents were considered in need of treatment for their opioid use if they met criteria for OUD. As defined by NSDUH (Center for Behavioral Health Statistics and Quality, 2016), participants were considered to have OUD if they met the DSM-IV criteria for opioid abuse or opioid dependence. Respondents were classified as having opioid dependence if in the past 12 months they met at least three of the seven DSM-IV criteria for dependence. Respondents were classified as having abused opioids if in the past 12 months they met at least one of the four DSM-IV criteria for abuse (American Psychiatric Association, 2000). Due to the DSM-IV's hierarchical decision rule that a diagnosis of dependence precludes abuse, respondents in the sample were coded as having either opioid abuse or dependence, but not both.

Respondents were excluded from the sample if they were < 18 years of age ($n = 86,393$), reported never having used an opioid recreationally ($n = 158,522$), did not meet the criteria for past year OUD ($n = 33,933$), or were missing data on a potential correlate of interest ($n = 73$). The final sample included 2488 respondents, representing 1,929,027 adults (Fig. 1).

2.3. Outcomes

The primary outcome of interest was receipt of specialty substance use treatment. Respondents were considered to have received specialty treatment if they reported receiving treatment or counseling for any drug use at a specialty treatment facility in the past 12 months. We considered the following to be specialty treatment facilities: rehabilitation facilities (residential or outpatient), hospitals (inpatient overnight services only), outpatient mental health centers, private doctor's office (for drug use), or methadone clinics. We focused on these

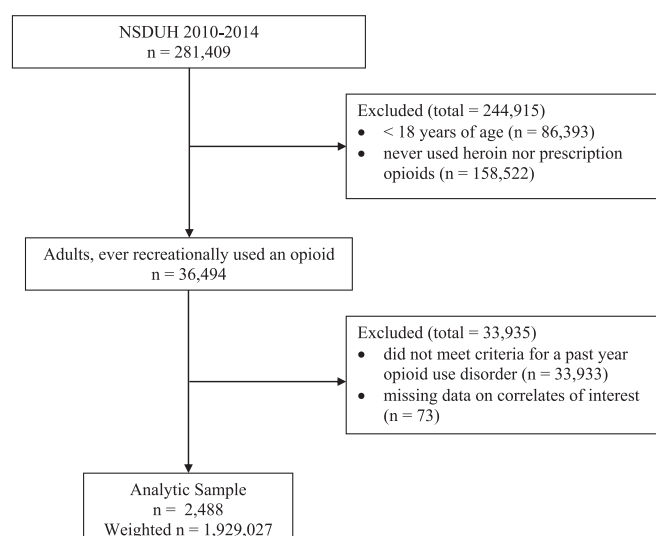


Fig. 1. Flow diagram of study sample.

facilities because these are locations where individuals could potentially receive evidence-based treatment for their drug use (e.g. methadone or buprenorphine). Given the effectiveness of detoxification alone as treatment has been questioned, we did not consider those who only received detoxification at one of these facilities as treated (Mattick & Hall, 1996). NSDUH does not specify whether drug treatment received in the past 12 months was specifically for an individual's OUD. However, given treatment guidelines recommending that individuals presenting with polysubstance use receive treatment that addresses each of their SUDs (Kleber et al., 2007; The Management of Substance Use Disorders, 2015), we believe an individual's OUD would be addressed at least to some extent, even if treatment was originally sought for a different SUD.

The secondary outcome was perceived need of treatment. Consistent with previous studies, respondents who did not report receiving treatment were considered to have a perceived need of treatment if they reported needing any treatment or counseling for their opioid use during the same time period (Hedden & Gfroerer, 2011; Mulvaney-Day, DeAngelo, Chen, Cook, & Alegría, 2012; Wu et al., 2011). Respondents who reported receiving treatment in the past 12 months were considered to have a perceived need of treatment if they reported that they needed *additional* treatment or counseling for their opioid use (i.e. the treatment they received was insufficient) (Hedden & Gfroerer, 2011; Mulvaney-Day et al., 2012; Wu et al., 2011). NSDUH does not differentiate what specific treatment respondents felt they needed (Center for Behavioral Health Statistics and Quality, 2014). Those with a perceived need of treatment were asked their reasons for not getting treatment/additional treatment.

2.4. Potential correlates

We applied the Gelberg-Andersen behavioral model for vulnerable populations when considering factors to evaluate as correlates of receiving specialty substance use treatment. This model outlines predisposing, enabling, and need-related factors that are considered to affect the use of health services, with an emphasis on factors that are particularly salient for vulnerable populations (e.g. substance users) (Gelberg, Andersen, & Leake, 2000).

Predisposing factors are sociodemographic factors that may predispose an individual to use health services (Andersen, 1995). The following were assessed as predisposing factors: age, gender, race/ethnicity, education, employment, marital status, household size, and criminal history.

Enabling factors are financial and contextual factors that may create

conditions that enable health service utilization (Andersen, 1995). The following were assessed as enabling factors: health insurance status, annual family income, receipt of public assistance, and the population density of a respondent's county of residence.

Need-related factors are factors that may affect an individual's need for health services (Andersen, 1995). The following were assessed as need-related factors: opioid use disorder (dependence vs. abuse), age of first opioid use, past-year alcohol use disorder (AUD); other past-year drug disorder, past-year mental, and self-reported health status.

2.5. Statistical analysis

Twelve-month prevalence estimates of the receipt of specialty substance use treatment were calculated using frequency distributions. Among those who received specialty treatment, frequency distributions were used to identify which facilities were the most common sources of treatment. Univariate logistic regression analysis was used to examine the crude association between each potential correlate and receipt of specialty substance use treatment. We then fit four multivariable logistic regression models to identify which variables in a given category were most associated with the receipt of specialty treatment and produce adjusted measures of association. The first model included all predisposing factors, the second all enabling factors, and the third all need-related factors. Finally, we fit a full model including all factors of interest. To obtain nationally representative estimates, sampling weights were applied to all analyses to account for NSDUH's complex sampling design. The design weights for NSDUH respondents adjust for differential selection probabilities, non-response, and make adjustments such that the distribution of the characteristics in the weighted sample are similar to that in the U.S. Census. All analyses were performed using Stata version 14. All reported results are weighted, with the exception of unweighted sample sizes. Twelve-month prevalence of perceived need of treatment and the prevalence of the most commonly reported reasons for not receiving treatment were estimated using frequency distributions. Our small sample size prevented stratifying our analysis of potential correlates by type of substance used (i.e. heroin use disorder or prescription opioid use disorder) and identifying correlates of perceived need of treatment.

3. Results

Of individuals with OUD, 8.3% received specialty treatment in the past 12 months (95% confidence interval (CI): 6.5% to 10.5%). Among those who received specialty treatment, 66.4% received treatment at 2 or more different specialty treatment facilities. Approximately one quarter (25.7%) of all respondents who received specialty treatment received treatment at 4 or more different specialty treatment facilities. As shown in Table 1, rehabilitation facilities were the most common treatment locations. Sixty-one percent of treated individuals received treatment at a residential rehabilitation facility and 61% received

Table 1
Location of specialty treatment received, among adults with an opioid use disorder who received specialty treatment in the last 12 months (n = 179, weighted n = 159,753).

Type of specialty facility	Unweighted n ^a	Weighted % ^b
Hospital, inpatient	71	55.0
Residential rehabilitation facility	98	60.6
Outpatient rehabilitation facility	98	60.6
Outpatient mental health facility	62	28.3
Private doctor's office	41	27.8
Methadone clinic	29	12.3

^a Survey weights were applied when calculating percentages. Therefore, weighted percentages differ from unweighted percentages.

^b Percentages do not add up to 100% because some respondents received treatment at more than one type of specialty treatment facility.

Table 2
Sample characteristics by specialty treatment status, weighted %.

Characteristic	Total	Received specialty treatment ^a	No specialty treatment	P-value
Sample N	2488	179	2309	
Weighted N	1,929,027	159,753	1,769,274	
Weighted percentages				
Predisposing factors				
Age (years)				
18–25	31.9	26.0	32.5	–
26–34	31.0	35.8	30.5	0.14
≥ 35	37.1	38.3	37.0	0.31
Men	60.5	57.0	60.8	0.54
Race/ethnicity				
Non-Hispanic White	72.2	70.5	72.3	–
Black/African American	9.9	10.2	9.9	0.90
Hispanic	12.7	15.2	12.5	0.61
Other	5.2	4.2	5.3	0.75
Education				
Less than high school	23.3	26.5	23.0	0.29
High school graduate	34.6	35.5	34.5	0.37
Some college	30.9	30.5	30.9	0.81
College graduate	11.3	7.5	11.6	–
Employment				
Employed full time	41.4	24.8	42.9	–
Employed part time	16.2	17.1	16.1	0.12
Unemployed	14.8	19.2	14.4	0.009
Not in the labor force ^b	27.7	38.9	26.7	0.003
Marital status				
Married	24.1	10.1	25.4	–
Widowed, divorced, or separated	19.1	20.9	18.9	0.01
Never been married	56.8	69.0	55.7	0.002
Number in household				
1 person (respondent)	9.3	13.2	8.9	–
2–3 people	47.6	34.3	48.8	0.15
4 or more people	43.1	52.5	42.3	0.74
Criminal history				
Never arrested/booked	40.6	15.9	42.9	–
Arrested/booked over 12 months ago	36.2	29.9	36.7	0.029
Arrested/booked in past 12 months	23.2	54.2	20.4	< 0.001
Enabling factors				
Health insurance				
None	30.3	27.7	30.5	0.18
Public	32.8	48.9	31.4	0.001
Private	36.9	23.4	38.1	–
Family income				
Less than \$20,000	31.8	45.5	30.5	0.008
\$20,000–\$49,000	35.8	34.8	35.9	0.22
\$50,000–\$74,999	12.7	7.2	13.2	0.76
\$75,000 or more	19.7	12.6	20.3	–
Receive public assistance	43.2	65.3	41.2	0.002
Population density				
Non-metro	15.0	6.7	15.7	–
Small metro	31.2	34.0	30.9	0.97
Large metro	53.9	59.3	53.4	0.003
Need-related factors				
Opioid use disorder				
Opioid abuse	19.6	6.7	20.8	–
Opioid dependence	80.4	93.3	79.2	< 0.001
Age of first use of opioid (years)				
≤ 12	7.6	8.2	7.6	0.88
13–17	31.0	34.3	30.7	0.75
18–25	41.4	37.6	41.7	0.79
≥ 26	20.0	19.9	20.1	–
Alcohol use disorder	36.0	60.3	33.9	< 0.001
Other drug use disorder	39.5	74.9	36.3	< 0.001
Mental illness				
No mental illness	39.9	24.3	41.3	–
Mild mental illness	19.9	17.4	20.1	0.32

Table 2 (continued)

Characteristic	Total	Received specialty treatment ^a	No specialty treatment	P-value
Moderate mental illness	16.8	14.8	16.9	0.28
Serious mental illness	23.5	43.5	21.7	< 0.001
Overall health				
Excellent/very good	42.9	42.6	42.9	–
Good	33.1	41.5	32.3	0.27
Fair/poor	24.0	15.9	24.7	0.22

^a Specialty treatment is defined as receiving treatment at any of the following locations: hospital (inpatient), residential rehabilitation facility, outpatient rehabilitation facility, outpatient mental health facility, private doctor's office, or methadone clinic.

^b includes individuals who are disabled, keeping house full-time, in school/training, retired, or do not have a job for some other reason.

treatment at an outpatient rehabilitation facility. Hospitals were the next most common treatment location, as 55% of treated individuals received treatment at a hospital overnight as an inpatient.

Of those who received specialty treatment in the past 12 months, 4.8% reported feeling a need for additional treatment for their opioid use. Among those who did not receive specialty treatment in the past 12 months, 9.7% reported feeling a need for treatment for their opioid use. The most commonly reported reason for not receiving treatment among those who perceived a need for treatment but did not receive any was not having health coverage and being unable to cover the cost of treatment (46.7%).

Table 2 displays predisposing, enabling, and need-related factors by specialty treatment status. The majority of adults with OUD were men, non-Hispanic White, and never married. With regard to predisposing factors, the treatment groups differed on employment status, marital status, and criminal history ($p < 0.05$). Among those reporting specialty treatment, 84.1% indicated a criminal history whereas 57.1% of those without specialty treatment had a criminal history. The proportion of those who had an arrest in the past 12 months was higher among those receiving specialty treatment compared to those who did not receive treatment (54.2% vs. 20.4%). The two groups differed on all enabling factors and on all need-related factors except age of first opioid use and overall health ($p < 0.05$). Approximately one-third of those who did not receive specialty treatment had co-occurring AUD (33.9%) or some other drug use disorder (36.3%). Both AUD and other drug use disorders were prevalent among those receiving specialty treatment (60.3% and 74.9%, respectively). More individuals reporting receiving treatment had a serious mental illness than did those without treatment (43.5% vs 21.7%, respectively).

Table 3 shows the association between each potential factor and receipt of specialty treatment. From the model including only predisposing factors, recent arrest (adjusted odds ratio (aOR): 7.09; 95% CI: 3.76–13.38) was more strongly associated with specialty treatment receipt relative to other predisposing factors examined. In the model including only enabling factors, living in a non-metro area was inversely associated with specialty treatment receipt (aOR: 0.32; 95% CI: 0.17–0.61). In the model including need factors, opioid dependence, AUD, and having another drug use disorder were all associated with increased odds of receiving specialty treatment. Each of these associations persisted even after adjusting simultaneously for pre-disposing, enabling, and need-related factors.

4. Discussion

In a large, nationally representative sample of adults with OUD in the U.S., we found that the prevalence of specialty substance use treatment receipt was low, as was the perceived need of treatment. Factors positively associated with past-year receipt of specialty substance use treatment included older age, being unemployed or out of

Table 3Association between different factors and receipt of specialty treatment^a, odds ratio (OR), 95% confidence interval (CI).

Variable	Crude OR (95% CI)	Adjusted OR			
		Predisposing factors adjusted OR (95% CI)	Enabling factors adjusted OR (95% CI)	Need factors adjusted OR 95% CI)	All factors adjusted OR (95% CI)
Predisposing factors					
Age (years)					
18–25	1.00	1.00			1.00
26–34	1.47 (0.88–2.43)	1.78 (1.02–3.12)*			1.62 (0.90–2.92)
≥ 35	1.29 (0.78–2.14)	2.36 (1.11–5.01)*			2.55 (1.04–6.26)*
Men	0.85 (0.52–1.41)	0.67 (0.43–1.06)			0.76 (0.50–1.16)
Race/ethnicity					
Non-Hispanic White	1.00	1.00			1.00
Black/African American	1.06 (0.44–2.53)	0.87 (0.36–2.11)			0.94 (0.37–2.40)
Hispanic	1.25 (0.52–2.97)	1.23 (0.53–2.90)			1.02 (0.48–2.17)
Other	0.82 (0.23–2.92)	1.18 (0.36–3.93)			0.92 (0.30–2.80)
Education					
College graduate	1.00	1.00			1.00
Less than high school	1.77 (0.61–5.15)	0.83 (0.26–2.64)			0.97 (0.35–2.72)
High school graduate	1.59 (0.57–4.41)	0.91 (0.31–2.63)			1.03 (0.41–2.59)
Some college	1.52 (0.55–4.24)	0.86 (0.30–2.48)			0.84 (0.35–2.05)
Employment					
Employed full time	1.00	1.00			1.00
Employed part time	1.84 (0.85–3.96)	1.49 (0.67–3.32)			1.36 (0.66–2.83)
Unemployed	2.32 (1.24–4.34)*	1.71 (0.87–3.37)			1.92 (1.02–3.61)*
Not in the labor force ^b	2.52 (1.39–4.55)*	2.08 (1.13–3.80)*			2.16 (1.15–4.06)*
Marital status					
Married	1.00	1.00			1.00
Widowed, divorced, or separated	2.78 (1.28–6.06)*	1.77 (0.78–4.00)			1.38 (0.62–3.06)
Never been married	3.12 (1.55–6.27)*	3.34 (1.65–6.74)*			2.14 (1.04–4.39)*
Number in household					
1 person (respondent)	1.00	1.00			1.00
2–3 people	0.48 (0.17–1.31)	0.56 (0.21–1.49)			0.56 (0.21–1.52)
4 or more people	0.84 (0.31–2.29)	0.98 (0.36–2.72)			0.87 (0.30–2.50)
Criminal history					
Never arrested/booked	1.00	1.00			1.00
Arrested over 12 months ago	2.20 (1.08–4.45)*	2.14 (1.03–4.43)*			1.71 (0.88–3.33)
Arrested in past 12 months	7.17 (3.81–13.50)*	7.09 (3.76–13.38)*			4.43 (2.45–7.99)*
Enabling factors					
Health insurance					
Private	1.00		1.00		1.00
Public	2.54 (1.46–4.40)*		1.52 (0.88–2.60)		1.05 (0.55–2.01)
None	1.48 (0.84–2.62)		1.06 (0.58–1.95)		0.67 (0.34–1.32)
Family income					
\$75,000 or more	1.00		1.00		1.00
Less than \$20,000	2.41 (1.27–4.59)*		1.43 (0.68–3.03)		1.12 (0.48–2.59)
\$20,000–\$49,000	1.57 (0.77–3.20)		1.19 (0.53–2.65)		1.13 (0.50–2.53)
\$50,000–\$74,999	0.88 (0.39–2.00)		0.84 (0.35–1.98)		0.72 (0.24–2.12)
Receive public assistance	2.68 (1.47–4.90)*		2.20 (1.16–4.20)*		1.73 (0.86–3.47)
Population density					
Large metro	1.00		1.00		1.00
Non-metro	0.38 (0.20–0.71)		0.32 (0.17–0.61)*		0.29 (0.12–0.68)*
Small metro	0.99 (0.56–1.75)		0.88 (0.50–1.52)		0.95 (0.55–1.63)
Need-related factors					
Opioid use disorder					
Opioid abuse	1.00			1.00	1.00
Opioid dependence	3.67 (2.01–6.71)*			3.89 (2.05–7.39)*	3.82 (2.06–7.10)*
Age of first use of opioids (years)					
≥ 26	1.00			1.00	1.00
≤ 12	1.09 (0.33–3.60)			0.87 (0.26–2.91)	0.80 (0.26–2.45)
13–17	1.13 (0.55–2.31)			0.66 (0.32–1.34)	0.91 (0.40–2.07)
18–25	0.91 (0.46–1.81)			0.63 (0.34–1.20)	0.77 (0.39–1.54)
Alcohol use disorder	2.97 (1.86–4.73)*			2.21 (1.39–3.52)	2.44 (1.44–4.11)*
Other drug use disorder	5.25 (3.13–8.82)*			4.41 (2.42–8.04)	3.22 (1.95–5.32)*
Mental illness					
No mental illness	1.00			1.00	1.00
Mild mental illness	1.47 (0.69–3.17)			1.14 (0.54–2.40)	1.13 (0.58–2.21)
Moderate mental illness	1.49 (0.72–3.07)			1.20 (0.57–2.53)	1.08 (0.57–2.07)
Serious mental illness	3.40 (1.91–6.02)*			2.07 (1.16–3.70)*	1.55 (0.85–2.82)
Overall health					
Excellent/very good	1.00			1.00	1.00
Good	1.29 (0.82–2.04)			1.29 (0.80–2.06)	1.21 (0.77–1.92)
Fair/poor	0.65 (0.32–1.31)*			0.56 (0.26–1.20)*	0.38 (0.17–0.86)*

^a Specialty treatment is defined as receiving treatment at any of the following locations: hospital (inpatient), residential rehabilitation facility, outpatient mental health facility, private

doctor's office, or methadone clinic.

^b includes individuals who are disabled, keeping house full-time, in school/training, retired, or do not have a job for some other reason.

^c $p < 0.05$.

the labor force, having never been married, an arrest in the past 12 months, opioid dependence (vs. abuse), having an AUD, and having another drug use disorder. Living in a non-metropolitan area and fair/poor self-reported overall health were both negatively associated with past-year receipt of specialty treatment.

We found that most adults with OUD did not receive specialty substance use treatment. This is consistent with other studies of national U.S. data estimating that the large majority of persons with any SUD, any OUD, or a prescription opioid use disorder did not receive any substance use treatment in the past twelve months (Center for Behavioral Health Statistics and Quality, 2015; Huang et al., 2006; Saloner & Karthikeyan, 2015). Saloner and Karthikeyan (Saloner & Karthikeyan, 2015), using 2009–2013 NSDUH data, found that 21.5% of individuals with OUD received treatment in the past year, over twice the percentage in our study. This is likely explained by the fact that Saloner and Karthikeyan used a more expansive definition of treatment, including emergency departments, jails, and self-help groups in their definition. In any case, compared to countries such as the United Kingdom and Germany, the U.S. treats a much lower percentage of individuals with OUD. Nearly 60% of problem opioid users in the U.K. receive opioid substitution treatment, while in Germany the proportion is nearly 50% (European Monitoring Centre for Drugs and Drug Addiction, Annual Report 2012, 2012). This has largely been attributed to many European countries using a treatment model where most opioid substitution medication is provided through generalist physicians in an office-based setting (Merrill, 2002).

Insufficient treatment capacity, policy and regulatory barriers, treatment-related stigma, and individuals not perceiving a need for treatment all contribute to this large treatment gap (Hedden & Gfroerer, 2011; Olsen & Sharfstein, 2014; Volkow, Frieden, Hyde, & Cha, 2014). Although NSDUH data do not allow us to investigate all potential explanations, our results do support the argument that a lack of perceived need of treatment is a major barrier to treatment receipt. We found that < 10% of adults with an OUD who did not receive specialty substance use treatment perceived a need for treatment of their opioid use. Similarly low rates of perceived need for treatment have been observed previously among individuals with any SUD (Ali, Teich, & Mutter, 2015; Mojtatabi et al., 2014). These results reflect a need for strategies to increase individuals' recognition of OUD symptoms and their need for treatment. Some have advocated for an expanded use of motivational interviewing, a counseling approach that has been shown to increase treatment initiation and retention among individuals with SUD (Carroll et al., 2006; Carroll, Libby, Sheehan, & Hyland, 2001). Negative consequences of opioid abuse disorder may also serve as external sources of motivation to initiate treatment. Indeed, we found that an arrest in the past 12 months was positively associated with receiving specialty treatment. While this may reflect cases of court ordered treatments, it might also reflect individuals' own efforts to demonstrate to the court that they are seeking rehabilitation.

Our results also show that barriers to accessing treatment exist even when individuals do perceive they need treatment. Among adults with an OUD who perceived a need for treatment but did not receive any, the most commonly reported reason for not getting treatment was having no health insurance and being unable to afford the cost of treatment (47%). Additionally, 15% reported that their health coverage did not cover substance use treatment or did not cover the full cost. These findings reiterate the need to expand coverage of substance use treatment in the U.S., particularly by expanding the gains from the Patient Protection and Affordable Care Act of 2010 (ACA). The ACA not only extended health insurance to an estimated 20 million previously uninsured people, but also required small group and individual insurance plans to provide coverage for substance use disorder services starting in

2014 (the final year in our analysis). The ACA builds on the Mental Health Parity and Addiction Equity Act of 2008, which prohibits large group health plans from imposing financial requirements or treatment limits on behavioral health benefits that are more restrictive than those imposed on medical/surgical benefits. It will be important to protect these policies and evaluate whether they are translating into increased utilization of substance use treatment.

Consistent with previous studies of individuals with alcohol or drug use disorder (Cook & Alegría, 2011; Saunders et al., 2006; Weisner, Matzger, Tam, & Schmidt, 2002), we found that receipt of specialty substance use treatment was positively associated with factors indicative of a life in disequilibrium: opioid dependence (vs. abuse), AUD, another drug use disorder, arrest in the past 12 months, and unemployment. One possible explanation for these findings is that adults with OUD may only receive treatment when their lives reach a certain level of dysfunction. If so, these results echo the need to expand the delivery of early intervention services. One strategy might be to further expand the implementation of Screening, Brief Intervention, and Referral to Treatment (SBIRT), an evidence-based practice that can be used by healthcare professionals in a variety of settings to assess patients for risky substance use, engage them in a short counseling conversation, and refer them to additional treatment if needed (Del Boca, McRee, Vendetti, & Damon, 2017). It would be especially useful to implement SBIRT in settings where persons misusing opioids are likely to interact with health professionals, such as mental health facilities, emergency departments, outpatient physician offices, and needle exchange programs.

Our finding that past-year treatment is associated with a life in "disequilibrium" may also speak to the nature of OUD. Those who reported receiving treatment might have begun receiving treatment prior to the study period. Therefore, another possible explanation for our findings is that despite receiving specialty treatment, adults with OUD continue to experience challenges. Despite OUD being a chronic illness characterized by relapse and remission (McLellan, Lewis, O'Brien, & Kleber, 2000), the U.S. healthcare system generally treats SUDs as acute conditions that are curable with a relatively brief episode of treatment (Dennis & Scott, 2007; McLellan et al., 2000). Adapting treatment programs to a chronic care approach will not change the chronic, relapsing nature of OUD but could improve individuals' long-term treatment outcomes (Dennis & Scott, 2007).

Finally, our finding that living in a non-metropolitan county was negatively associated with receiving specialty substance use treatment is consistent with research that shows individuals with opioid dependence in rural areas face inadequate treatment capacity and long travel times (Sigmon, 2014). This is especially alarming given rates of non-medical use of prescription opioids and opioid-related deaths are higher in rural areas (Keyes, Cerdá, Brady, Havens, & Galea, 2014). Federal and state agencies are currently funding a number of projects targeting treatment in rural communities (RFA-HS-16-001, 2017; Rural Health Opioid Program, 2017; State-Targeted-Response-to-Opioid-Crisis-Grant, 2017). As these projects bear fruit, it will be important to ensure that efforts are made to translate evidence into policy.

This study is not without limitations. Although we evaluated whether adults with OUD received specialty substance use treatment, we could not determine whether the treatment received was specifically for an individual's OUD. There are also limitations to our definition of specialty treatment facilities. Some jails and emergency departments, which we did not consider specialty treatment facilities, offer extensive substance use treatment programs and SBIRT, respectively. Also, our exclusion of self-help groups may have caused us to erroneously classify as untreated the subset of individuals who may be exclusively attending self-help groups for relapse prevention. In both situations, we expect

the number of treated individuals missed to be small. It is also worth noting that some treatments, including MAT, are typically only recommended for individuals with opioid dependence. Therefore, our finding that individuals with opioid dependence are more likely to receive treatment than those with opioid abuse may be explained by their access to a wider range of treatment options. The NSDUH also has limitations. Respondents may have selectively participated. Although the sampling weights attempt to account for such selection biases, the extent to which NSDUH respondents adequately represent all adults with OUD remains unknown. Also, there is potential for inaccurately recalled information related to their substance use, receipt of treatment, or potential correlates of interest. If those who received specialty treatment also were more likely to accurately recall their opioid use severity or other drug use, this could explain our findings.

Our study has a number of important strengths. First, the NSDUH offers rich information on substance use and treatment utilization for one of the largest available samples of substance users. Our study applied sampling weights thereby allowing for the calculation of nationally representative estimates. NSDUH employs several strategies to reduce recall bias concerns such as a relatively short recall period of 12 months. Additionally, NSDUH's use of ACASI to collect sensitive information may have mitigated socially desirable responding, as has been observed in other contexts (Turner et al., 1998).

The current U.S. opioid crisis is a public health epidemic in need of significant action. Our study reiterates the large gap between need and treatment for OUD. Efforts at multiple levels must be made to identify and provide appropriate interventions for both populations at risk of OUD as well as those in the early stages of opioid misuse. Also, efforts to increase treatment capacity for OUD should be monitored to ensure that populations with the most need are prioritized.

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Contributors

All authors have contributed to and approved the final manuscript. Eric Romo and Kate Lapane designed the initial study and protocol. Eric Romo performed literature searches, conducted the analyses, and wrote the first draft of the manuscript. Christine Ulbricht, Robin Clark, and Kate Lapane provided input throughout the conduct of the study and edited subsequent versions of the manuscript.

Conflict of interest

There are no conflicts of interest.

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